

June 17, 2026

Dear PhD Admissions Committee,

I am excited to apply for the PhD position in Quantum Reservoir Computing via Nano-Opto-Electro-Mechanical (NOEM) coupled quantum oscillators at the University of Southampton. Located in Southampton, United Kingdom, this esteemed institution has a strong reputation for fostering innovation and excellence.

As an AI Data Science Engineer with a passion for applying mathematical models to real-world problems, I am particularly drawn to this project's focus on developing novel hardware for quantum neural networks. My research experience in computer science, physics, and electronic engineering has equipped me with a solid foundation to tackle the challenges of this project.

One achievement that stands out as highly relevant is my development of *MedicalMambaTS*, a linear-time state space model capable of generating sub-millisecond bedside inference for cardiovascular medical ap

Another achievement that aligns perfectly with this project is *PIDeepONet*, a 0D Windkessel Physics-Informed Operator Network using PyTorch Autograd. This work demonstrated my ability to leverage physical

Sincerely,

Ismail Aissaoui